

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — SOLUTIONS & MARKING SCHEME

Subject: Differential Equation & Linear Algebra

Time: 1.5 Hours

Semester / Programme: First Semester

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

1(a) Find the integrating factor of the first-order linear differential equation $x \frac{dy}{dx} + yx = 1$. [1]

$$I.F. = e^{\int P(x) dx}$$

Solution & Key Concept:

- Rewrite the given equation in standard form $(\frac{dy}{dx}) + P(x)y = Q(x)$ by dividing throughout by x : $(\frac{dy}{dx}) + (x)y = \frac{1}{x}$.
- Identify $P(x) = x$.
- Calculate the integrating factor: $\{I.F.\} = e^{\int x dx} = e^{\frac{1}{2}x^2} = x$.

Answer: $\sec x$

Marking Rubric: [0.5 Mark] for rewriting in standard form, [0.5 Mark] for correct integrating factor.

1(b) Determine the Wronskian $W(y_1, y_2)$ of the two solutions $y_1 = e^{2x}$ and $y_2 = e^{-3x}$ of a second-order linear differential equation. [1]

$$\text{Formula: } W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$$

Solution & Key Concept:

- Compute derivatives: $y_1' = 2e^{2x}$ and $y_2' = -3e^{-3x}$.
- Evaluate the determinant: $W = \begin{vmatrix} e^{2x} & e^{-3x} \\ 2e^{2x} & -3e^{-3x} \end{vmatrix}$.
- $W = e^{2x}(-3e^{-3x}) - (e^{-3x})(2e^{2x}) = -3e^{-x} - 2e^{-x} = -5e^{-x}$.

Answer: $-5e^{-x}$

Marking Rubric: [0.5 Mark] for setting up Wronskian determinant, [0.5 Mark] for final simplification.

1(c) State the condition for exactness of a first-order ordinary differential equation $M(x,y)dx + N(x,y)dy = 0$. [1]

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Solution & Key Concept:

- For a first-order differential equation $M(x,y)dx + N(x,y)dy = 0$ to be exact in a simply connected region, the partial derivatives of M with respect to y and N with respect to x must be continuous and equal everywhere in that region.

Answer: $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Marking Rubric: [1 Mark] for stating the exactness condition correctly.

1(d) Find the recurrence relation for the Gamma function $\Gamma(x)$ for $x > 0$. [1]

$$\Gamma(x+1) = x\Gamma(x)$$

Solution & Key Concept:

- Recall definition: $\Gamma(x) = \int_0^\infty e^{-t} t^{x-1} dt$.
- Using integration by parts: $\Gamma(x+1) = \int_0^\infty e^{-t} t^x dt = \left[-t^x e^{-t} \right]_0^\infty + \int_0^\infty x t^{x-1} e^{-t} dt = 0 + x\Gamma(x)$.

Answer: $\Gamma(x+1) = x\Gamma(x)$

Marking Rubric: [1 Mark] for correct recurrence relation formula.

1(e) Write the auxiliary (characteristic) equation for the second-order differential equation $y'' - 5y' + 6y = 0$.

[1]

$$ar^2 + br + c = 0$$

Solution & Key Concept:

- Replace y'' with r^2 , y' with r , and y with 1 in the homogeneous linear differential equation with constant coefficients.
- Obtain $r^2 - 5r + 6 = 0$.

Answer: $r^2 - 5r + 6 = 0$

Marking Rubric: [1 Mark] for writing the correct auxiliary quadratic equation.

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

Q.2. (a) Solve the Bernoulli differential equation: $(dy/dx) + (1/x)y = xy^2$.

[2.5]

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

Solution for (a):

- Divide the entire equation by y^2 : $y^{-2}(dy/dx) + (1/x)y^{-1} = x$.
- Substitute $v = y^{-1} = (1/y)$. Then $(dv/dx) = -y^{-2}(dy/dx)$, or $y^{-2}(dy/dx) = -(dv/dx)$.
- Substitute into the equation to get a linear equation in v : $-(dv/dx) + (1/x)v = x$ $(dv/dx) - (1/x)v = -x$.
- Find Integrating Factor: $\{I.F.\} = e^{\int \{-1/x\} dx} = e^{-\ln x} = (1/x)$.
- Multiply by I.F. and integrate: $(d/dx) \leq f((v/x)) = -1 (v/x) = -x + C$ $v = -x^2 + Cx$.
- Substitute back $v = (1/y)$: $(1/y) = x(C - x)$.

Result / Conclusion: $y = \frac{1}{x(C - x)}$

Marking Rubric: [1 Mark] for substitution and transforming to linear form, [1.0 Mark] for solving the linear differential equation in v , [0.5 Mark] for back-substitution and final general solution.

(b) A thermometer reading 70°F is placed in an oven preheated to a constant temperature. Through Newton's Law of Cooling/Heating, if the thermometer reads 110°F after 1 minute and 140°F after 2 minutes, what is the oven temperature?

[2.5]

$$\frac{dT}{dt} = -k(T - T_{\text{oven}})$$

Solution for (b):

- Let $T_{\text{oven}} = T_m$ be the constant temperature of the oven. By Newton's Law: $(dT/dt) = -k(T - T_m)$.
- Separating variables and integrating gives: $T(t) = T_m + (T_0 - T_m)e^{-kt}$.
- Given initial temperature $T(0) = 70$: $T(t) = T_m + (70 - T_m)e^{-kt}$.
- At $t = 1$, $T(1) = 110$ $110 = T_m + (70 - T_m)e^{-k}$ $110 - T_m = (70 - T_m)e^{-k}$ -- (1).
- At $t = 2$, $T(2) = 140$ $140 = T_m + (70 - T_m)e^{-2k}$ $140 - T_m = (70 - T_m)(e^{-k})^2$ -- (2).
- Substitute (e^{-k}) from (1) into (2): $140 - T_m = (70 - T_m) \leq f((110 - T_m / 70 - T_m))^2 = ((110 - T_m)^2 / 70 - T_m)$.
- $(140 - T_m)(70 - T_m) = (110 - T_m)^2$ $9800 - 210T_m + T_m^2 = 12100 - 220T_m + T_m^2$.
- Simplify: $10T_m = 12100 - 9800 = 2300$ $T_m = 230^\circ\text{F}$.

Result / Conclusion: $T_{\text{oven}} = 230^\circ\text{F}$

Marking Rubric: [1.0 Mark] for setting up the differential equation and general solution, [1.0 Mark] for algebraic manipulation using data points at $t=1$ and $t=2$, [0.5 Mark] for correct oven temperature evaluation.

Q.3. (a) Solve the exact differential equation: $(3x^{2y} + y^3)dx + (x^3 + 3xy^2)dy = 0$. [2.5]

$$\int_{y=\text{const}} M dx + \int (\text{terms of } N \text{ not containing } x) dy = C$$

Solution for (a):

- Identify $M = 3x^{2y} + y^3$ and $N = x^3 + 3xy^2$.
- Check exactness: $(M/y) = 3x^2 + 3y^2$ and $(N/x) = 3x^2 + 3y^2$. Since $(M/y) = (N/x)$, the equation is exact.
- Integrate M with respect to x treating y as constant: $\int (3x^{2y} + y^3) dx = x^{3y} + xy^3$.
- Integrate terms of N free of x with respect to y : $\int 0 \cdot dy = 0$.
- Combine results to write the general solution: $x^{3y} + xy^3 = C$.

Result / Conclusion: $x^{3y} + xy^3 = C$

Marking Rubric: [0.5 Mark] for exactness verification, [1.5 Marks] for correct integration of M and N , [0.5 Mark] for general solution.

(b) Find the orthogonal trajectories of the family of curves $y = Cx^2$, where C is an arbitrary parameter. [2.5]

$$\text{Formula: } \left| \frac{dy}{dx} \right|_{\text{orthogonal}} = - \frac{1}{\left| \frac{dy}{dx} \right|}$$

Solution for (b):

- Differentiate the given family $y = Cx^2$ implicitly with respect to x : $(dy/dx) = 2Cx$.
- Eliminate the parameter C from the derivative expression. From original equation, $C = (y/x^2)$.
- Substitute C : $(dy/dx) = 2 \leq f((y/x^2))x = (2y/x)$.
- Replace (dy/dx) with $-(dx/dy)$ to get the differential equation for the orthogonal trajectories: $-(dx/dy) = (2y/x)$ $x dx + 2y dy = 0$.
- Integrate both sides: $\int x dx + \int 2y dy = \int 0 \Rightarrow (x^2/2) + y^2 = K_1 \Rightarrow x^2 + 2y^2 = K$.

Result / Conclusion: $x^2 + 2y^2 = K$

Marking Rubric: [1.0 Mark] for finding the slope of the original family and eliminating the parameter, [1.0 Mark] for setting up the orthogonal differential equation, [0.5 Mark] for integrating to get the final family of curves.

Q.4. (a) Solve the second-order homogeneous differential equation with constant coefficients: $y'' - 4y' + 4y = 0$. [2.5]

$$y(x) = (C_1 + C_2 x)e^{rx}$$

Solution for (a):

- Write down the auxiliary equation: $r^2 - 4r + 4 = 0$.
- Factor the quadratic equation: $(r - 2)^2 = 0$.
- Obtain the roots: repeated real roots $r_1 = r_2 = 2$.
- Write the general solution for repeated roots: $y(x) = (C_1 + C_2 x)e^{2x}$.

Result / Conclusion: $y(x) = (C_1 + C_2 x)e^{2x}$

Marking Rubric: [1.0 Mark] for auxiliary equation and finding roots, [1.5 Marks] for correct general solution structure for repeated roots.

(b) Solve the Euler-Cauchy equation: $x^2 y'' - 3x y' + 4y = 0$. [2.5]

$$m(m-1) + \alpha m + \beta = 0$$

Solution for (b):

- Substitute $y = x^m$, $xy' = m x^m$, and $x^2 y'' = m(m-1)x^m$ into the differential equation.
- Factor out x^m to get the auxiliary polynomial equation in m : $m(m-1) - 3m + 4 = 0$.
- Simplify: $m^2 - m - 3m + 4 = 0 \Rightarrow m^2 - 4m + 4 = 0$.
- Factor: $(m - 2)^2 = 0 \Rightarrow m = 2, 2$ (repeated roots).
- Write the general solution for Euler-Cauchy equation with repeated roots: $y(x) = (C_1 + C_2 x)x^2$.

Result / Conclusion: $y(x) = (C_1 + C_2 \ln x)x^2$

Marking Rubric: [1.0 Mark] for auxiliary polynomial equation, [1.0 Mark] for solving roots, [0.5 Mark] for correct general solution format.

Q.5. (a) Evaluate the Beta integral representation relation and prove that $B(m, n) = ((m)(n) / (m+n))$ for $m, n > 0$. [2.5]

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

Solution for (a):

- Consider the product of two Gamma functions: $(m)(n) = \int_0^\infty \int_0^\infty e^{-u} u^{m-1} du \int_0^\infty e^{-v} v^{n-1} dv$.
- Transform to double integrals: $(m)(n) = \int_0^\infty \int_0^\infty e^{-(u+v)} u^{m-1} v^{n-1} du dv$.
- Substitute polar/scaling coordinates: Let $u = x^2$ and $v = y^2$ or use substitution $u = zx, v = z(1-x)$. Let $u+v = z$ and $u = zx$, so $du = z dx$ and $dv = z(1-x)$.
- The limits for z are 0 to ∞ and for x are 0 to 1.
- $(m)(n) = \int_0^\infty \int_0^1 e^{-z} z^{m+n-1} dx dz = \int_0^\infty x^{m-1} (1-x)^{n-1} dx = (m+n) B(m, n)$.
- Thus, $B(m, n) = ((m)(n) / (m+n))$.

Result / Conclusion: $B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$

Marking Rubric: [1.0 Mark] for setting up the product of Gamma functions, [1.0 Mark] for coordinate transformation/separation, [0.5 Mark] for final relation derivation.

(b) Find the radius of convergence of the power series $\sum_{n=1}^\infty (3^n / n!) x^n$. [2.5]

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

Solution for (b):

- Identify the coefficient $a_n = (3^n / n!)$.
- Apply the ratio formula for the radius of convergence R : $(1/R) = \lim_{n \rightarrow \infty} |a_{n+1} / a_n|$.
- Compute $|a_{n+1} / a_n| = (3^{n+1} / (n+1)!) \cdot (n! / 3^n) = (3 / (n+1))$.
- Evaluate the limit as $n \rightarrow \infty$: $(1/R) = \lim_{n \rightarrow \infty} (3 / (n+1)) = 0$.
- Since $(1/R) = 0$, the radius of convergence $R = \infty$.

Result / Conclusion: $R = \infty$

Marking Rubric: [1.0 Mark] for identifying a_n and setting up D'Alembert's ratio test, [1.0 Mark] for evaluating the limit, [0.5 Mark] for stating radius of convergence as infinity.

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